

Chains of reasoning - Set 1

TMUA practice questions by topic. Specification Prf2, Prf3, Prf5. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1** In a knockout tournament with 37 players, each match eliminates exactly one player and the tournament ends when one player remains. How many matches are played?
- A** 36
B 35
C 19
D 37
E 64
F 18
- 2** The digits 1 to 9 are placed in a 3×3 grid so that every row and every column has the same sum. What is that sum?
- A** 45
B 15
C 20
D 18
E 12
F 10
- 3** Each of 100 cards has a different integer from 1 to 100 written on it. What is the smallest number of cards that must be drawn (without looking) to be certain of having two cards whose numbers sum to 101?
- A** 51
B 2
C 100
D 99
E 52
F 50

- 4 In a class every student studies at least one of Maths and Physics. 70% study Maths and 60% study Physics. A student is picked at random. What is the probability they study both?
- A 0.42
 - B 0.2
 - C 0.1
 - D 0.3
 - E 0.5
 - F it cannot be determined
- 5 The sum of 5 consecutive odd numbers is S . Which of the following is always a factor of S ?
- A 4
 - B 3
 - C 10
 - D 25
 - E 15
 - F 5
- 6 Seven consecutive integers are written down. Which of the following statements is/are true?
- I Their sum is divisible by 7.
 - II Their product is divisible by 7.
 - III Their sum is odd.
- A none of them
 - B I only
 - C II only
 - D III only
 - E I and II only
 - F I and III only
 - G II and III only
 - H I, II and III

- 7 A 4×4 grid of squares is coloured so that every row and every column contains exactly two black squares. How many black squares are there in the grid?
- A** 6
B 16
C 10
D 8
E 4
F 12
- 8 A selection of n distinct integers is taken from $1, 2, \dots, 20$. What is the smallest n that guarantees two of the chosen integers differ by exactly 10?
- A** 2
B 15
C 12
D 11
E 20
F 10
- 9 Five friends each shake hands with some of the others (each pair shakes hands at most once). Which of the following is impossible?
- A** Every friend shakes hands with exactly two others.
B Every friend shakes hands with exactly four others.
C The numbers of handshakes made are 1, 1, 2, 2, 2.
D The numbers of handshakes made by the five friends are 4, 4, 4, 4, 1.
E Exactly one friend shakes no hands at all.
- 10 x and y are positive integers with $x^2 - y^2 = 45$. How many possible pairs (x, y) are there?
- A** 0
B 6
C 2
D 1
E 3
F 4

11 Three statements are made about a positive integer N :

- N is a perfect square.
- N is a multiple of 6.
- N is less than 50.

Exactly two of the statements are true. Which of the following could NOT be N ?

- A** 144
- B** 12
- C** 36
- D** 48
- E** 16
- F** 25

12 a , b and c are real numbers with $a < b < c$. Which of the following statements must be true?

I $a + b < 2c$

II $ab < bc$

III $\frac{a+c}{2} < c$

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III